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# Project Description

Sep 17, 2026

## OVERVIEW

Time has come to propose and implement your project. Couple of guidelines on how to select a good project:

- Spend a good time thinking about a problem statement. You should not be picking a problem statement which is too ambitious such as creating a new operating system. You should also should not be picking something that is too simplistic such as printing Hello World
- You might want to do something that you don't yet know how to do. Such as
  - Creating a simple memory allocator
  - Rewriting Shell commands such as
    - cat
    - chmod
    - echo
  - Write a simple compiler
  - Write a simple scheduler
  - Write your own standard library etc.
  - Make a game in C from scratch
  - Implementing solitaire
  - Implement poker
  - Build your own shell
  - A regular expression matcher
  - Create your own text editor
  - Write your own libraries
    - Encoding and Decoding
    - Sorting
    - Searching etc..
  - Write BigInt in C
  - Build a simple database
  - Simple TCP Handshake
  - Or the one you like...

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## GOALS

1. Goal of giving the project is to let you to do critical thinking on selecting a problem statement, create a story on the project and then do the actual implementation
2. As a side benefit you will get a project ready to showcase as part of interviews. Keeping this in mind, we recommend that each of the student pick up their own problem statement

## DELIVERABLES

As part of the project, you will check into your github repository:

- A project proposal covering project description, goals, specifications and design. Project proposal needs to be part of the README.md file in the github repository. README.md needs to be written in Markdown language.
- Project code **needs to be written in C**. This is non-negotiable

## MILESTONES

- **Sep 17, 2026** - Completed project with all deliverables checked in github repository